



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

01-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE : ALL STAR BATCH)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A	C	C	D	D	D	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	C	D	A	A	D	D	A	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	7	30	543	14	60	2	3	120	50	15

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	C	A	A	C	C	A	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	B	D	B	C	B	C	D	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	42	182	38	5	109	50	5	269	0	3

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	C	B	C	C	C	D	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	C	C	A	D	C	A	C	A	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	7744	114	5	2021	96	49	18	12	4	6

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)

$$\begin{array}{ccc} X & Y & Z \\ m_1 = m & m_2 = m & m_3 = m \\ T_1 = 10^\circ\text{C} & T_2 = 20^\circ\text{C} & T_3 = 30^\circ\text{C} \\ s_1 & s_2 & s_3 \end{array}$$

when x & y are mixed, $T_{f1} = 16^\circ\text{C}$

$$m_1s_1T + m_2s_2T_2 = (m_1s_1 + m_2s_2)T_{f1}$$

$$s_1 \times 10 + s_2 \times 20 = (s_1 + s_2) \times 16$$

$$s_1 = \frac{2}{3}s_2 \dots\dots(i)$$

when y & z are mixed, $T_{f2} = 26^\circ\text{C}$

$$m_2s_2T + m_3s_3T_3 = (m_2s_2 + m_3s_3)T_{f2}$$

$$s_2 \times 20 + s_3 \times 30 = (s_2 + s_3) \times 26$$

$$s_3 = \frac{3}{2}s_2 \dots\dots(ii)$$

when x & z are mixed

$$m_1s_1T_1 + m_3s_3T_3 = (m_1s_1 + m_3s_3)T_f$$

$$\frac{2}{3}s_2 \times 10 + \frac{2}{3}s_2 \times 20 = \left(\frac{2}{3}s_2 + \frac{3}{2}s_2\right) T_f$$

$$T_f = 23.84^\circ\text{C}$$

2. Ans (B)

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}, mv = \sqrt{2mE}$$

$$\lambda \propto \frac{1}{\sqrt{E}}$$

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{E_1}{E_2}} = \frac{3}{4}, \lambda_2 = 0.75 \lambda_1$$

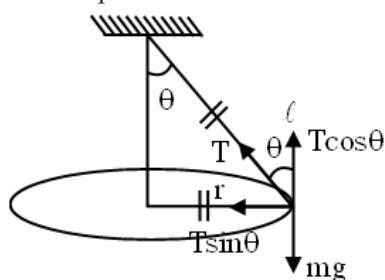
$$\frac{E_1}{E_2} = \left(\frac{3}{4}\right)^2$$

$$E_2 = \frac{16}{9}E_1 = \frac{16}{9}E \quad (E_1 = E)$$

$$\text{Extra energy given} = \frac{16}{9}E - E = \frac{7}{9}E$$

3. **Ans (A)**

Conical pendulum



$$r = \frac{l}{\sqrt{2}}$$

$$\sin \theta = \frac{r}{l} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

$$T \sin \theta = \frac{mv^2}{r}$$

$$T \cos \theta = mg$$

$$\tan \theta = \frac{v^2}{rg} \Rightarrow v = \sqrt{rg}$$

4. **Ans (C)**

$$V = 4 \times 10^{-3} \text{ m}^3$$

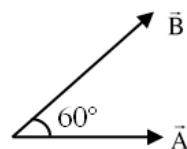
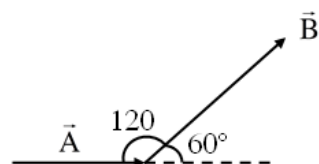
$$n = 3 \text{ moles}$$

$$T = 400 \text{ K}$$

$$PV = nRT \Rightarrow P = \frac{nRT}{V}$$

$$P = \frac{3 \times 8.3 \times 400}{4 \times 10^{-3}} = 24.9 \times 10^5 \text{ Pa}$$

5. **Ans (C)**



Angle between $\vec{A}$ and $\vec{B}$, $\theta = 60^\circ$

Angle between $\vec{A}$ and $\vec{A} - \vec{B}$

$$\tan \alpha = \frac{B \sin \theta}{A - B \cos \theta}$$

$$= \frac{B \sqrt{\frac{3}{2}}}{A - B \times \frac{1}{2}}$$

$$\tan \alpha = \frac{\sqrt{3}B}{2A - B}$$

6. **Ans (D)**

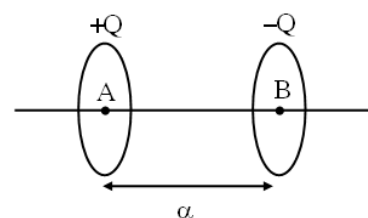
$$\frac{E_0}{C} = B_0$$

$$F_{\max} = eB_0V$$

$$= 1.6 \times 10^{-19} \times \frac{800}{3 \times 10^8} \times 3 \times 10^7$$

$$= 12.8 \times 10^{-18} \text{ N}$$

7. **Ans (D)**



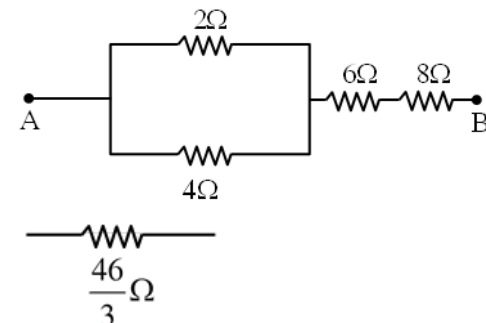
$$V_A = \frac{KQ}{a} - \frac{KQ}{\sqrt{a^2 + s^2}}$$

$$V_B = \frac{-KQ}{a} + \frac{KQ}{\sqrt{a^2 + s^2}}$$

$$V_A - V_B = \frac{2KQ}{a} - \frac{2KQ}{\sqrt{a^2 + s^2}}$$

$$= \frac{Q}{2\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{s^2 + a^2} \right)$$

8. **Ans (D)**



9. **Ans (D)**

Parallel axis theorem

$$I = I_{CM} + Md^2$$

$$I = \frac{Mr^2}{2} + M \left(\frac{L}{2} \right)^2$$

$$2.7 = M \frac{(0.2)^2}{2} + M \left(\frac{0.8}{2} \right)^2$$

$$2.7 = M \left[\frac{2}{100} + \frac{16}{100} \right]$$

$$M = 15 \text{ kg}$$

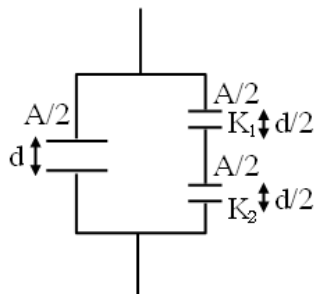
$$\Rightarrow \rho = \frac{M}{\pi r^2 L} = \frac{15}{\pi (0.2)^2 \times 0.8}$$

$$= 0.1492 \times 10^3$$

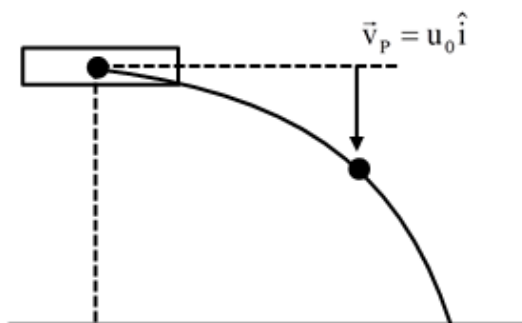
10. Ans (A)

$$C_{eq} = \frac{\frac{A}{2}\epsilon_0}{d} + \frac{A\epsilon_0}{d} \frac{K_1 K_2}{K_1 + K_2}$$

$$= \frac{A\epsilon_0}{d} \left(\frac{1}{2} + \frac{K_1 K_2}{K_1 + K_2} \right)$$



11. Ans (C)



$$v_B = u_0 \hat{i} - gt \hat{j}$$

$$\vec{v}_{B/P} = \vec{v}_B - \vec{v}_P$$

$$\vec{v}_{B/P} = -8t \hat{j}$$

straight line vertically down

12. Ans (C)

A $\rightarrow$ B, B $\rightarrow$ C

$$\frac{dN_B}{dt} = \lambda N_A - \lambda N_B$$

$$\frac{dN_B}{dt} = 2\lambda N_{B_0} e^{-\lambda t} - \lambda N_B$$

$$e^{-\lambda t} \left(\frac{dN_B}{dt} + \lambda N_B \right) = 2\lambda N_{B_0} e^{-\lambda t} \times e^{\lambda t}$$

$$\frac{d}{dt} (N_B e^{\lambda t}) = 2\lambda N_{B_0}, \text{ on integrating}$$

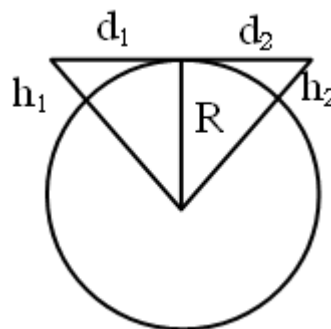
$$N_B e^{\lambda t} = 2\lambda t N_{B_0} + N_{B_0}$$

$$N_B = N_{B_0} [1 + 2\lambda t] e^{-\lambda t}$$

$$\frac{dN_B}{dt} = 0 \text{ at } -\lambda [1 + 2\lambda t] e^{-\lambda t} + 2\lambda e^{-\lambda t} = 0$$

$$N_{B_{max}} \text{ at } t = \frac{1}{2\lambda}$$

13. Ans (D)



$$d_t = \sqrt{2Rh_1} + \sqrt{2Rh_2}$$

$$= \sqrt{2R} (\sqrt{h_1} + \sqrt{h_2})$$

$$= (2 \times 6400 \times 10^3)^{1/2} (\sqrt{50} + \sqrt{80})$$

$$= 3578 (7.07 + 8.94)$$

$$= 57.28 \text{ Km}$$

14. Ans (A)

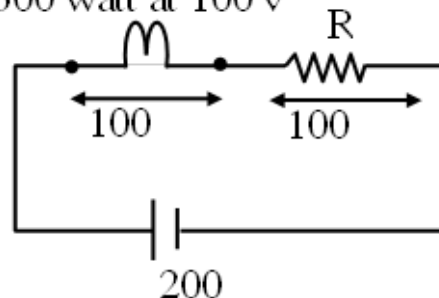
$$\frac{T_L}{T_H - T_L} = \text{C.O.P.} = \frac{\frac{dH}{dt}}{\frac{dW}{dt}}$$

$$\frac{263}{35} \times 35 = \frac{dH}{dt}$$

$$\frac{dH}{dt} = 263 \text{ watts}$$

15. Ans (A)

500 watt at 100 v



$$P = Vi$$

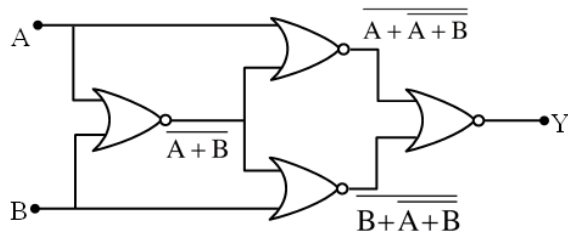
$$500 = Vi$$

$$i = 5 \text{ Amp}$$

$$V = i \times R$$

$$R = 20$$

16. Ans (D)



$$y = (A + \overline{A + B}) + (\overline{B + A + B})$$

$$y = (A + \overline{A + B}) \cdot (\overline{B + A + B})$$

A	B	y
0	0	1
0	1	0
1	0	0
1	1	1

17. Ans (D)

(a) Magnetic Induction = $MT^{-2}A^{-1}$

(b) Magnetic Flux = $ML^2T^{-2}A^{-1}$

(c) Magnetic Permeability = $MLT^{-2}A^{-2}$

(d) Magnetization = $M^0L^{-1}A$

18. Ans (A)

$$Z_C = \sqrt{\left(\frac{1}{\omega C}\right)^2 + R^2}$$

$$= \sqrt{\left(\frac{1}{100 \times 100 \times 10^{-6}}\right)^2 + 100^2}$$

$$Z_C = \sqrt{(100)^2 + (100)^2}$$

$$= 100\sqrt{2}$$

$$Z_L = \sqrt{(\omega L)^2 + R^2}$$

$$\sqrt{(100 \times 0.5)^2 + 50^2}$$

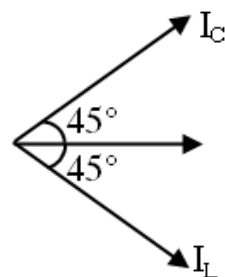
$$= 50\sqrt{2}$$

$$i_C = \frac{200}{Z_C} = \frac{200}{100\sqrt{2}} = \sqrt{2}$$

$$i_L = \frac{200}{Z_L} = \frac{200}{50\sqrt{2}} = 2\sqrt{2}$$

$$\cos \phi_1 = \frac{100}{10\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \phi_1 = 45^\circ$$

$$\cos \phi_2 = \frac{50}{50\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \phi_2 = 45^\circ$$



$$I = \sqrt{I_C^2 + I_L^2}$$

$$= \sqrt{2 + 8} = \sqrt{10}$$

$$I = 3.16 \text{ A}$$

Ans. 3.16

19. Ans (C)

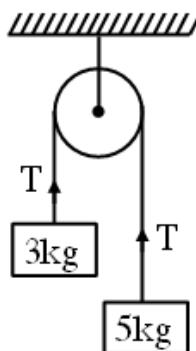
$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$\frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta \ell}{\ell}$$

$$\Delta T = \frac{1}{2} \times \frac{0.1}{100} \times 24 \times 3600$$

$$\Delta T = 43.2$$

20. Ans (C)



$$T = \frac{2m_1m_2g}{m_1 + m_2} = \frac{2 \times 3 \times 5 \times 10}{8}$$

$$= \frac{75}{2}$$

$$\text{Stress} = \frac{T}{A}$$

$$\frac{24}{\pi} \times 10^2 = \frac{75}{2 \times \pi R^2}$$

$$R^2 = \frac{75}{2 \times 24 \times 100} = \frac{3}{8 \times 24}$$

$$\Rightarrow R = 0.125 \text{ m}$$

$$R = 12.5 \text{ cm}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (7)**

$$y_1 = A_1 \sin k(x - vt)$$

$$y_1 = 12 \sin 6.28 (x - vt)$$

$$y_2 = 5 \sin 6.28 (x - vt + 3.5)$$

$$\Delta \phi = \frac{2\pi}{\lambda} (\Delta x)$$

$$= K(\Delta x)$$

$$= 6.28 \times 3.5 = \frac{7}{2} \times 2\pi = 7\pi$$

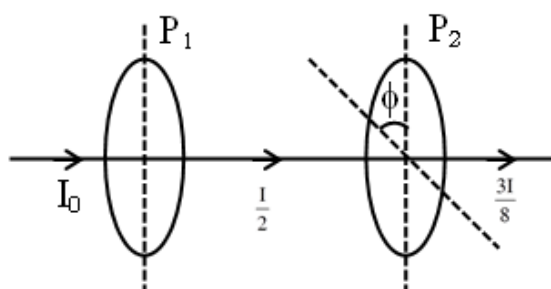
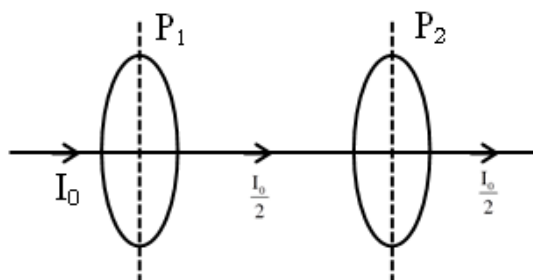
$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos \phi}$$

$$A_{\text{net}} = \sqrt{(12)^2 + (5)^2 + 2(12)(5) \cos(7\pi)}$$

$$= \sqrt{144 + 25 - 120}$$

2. **Ans (30)**

$$I = \frac{I_0}{2} \cos^2 \phi$$



$$\frac{I}{2} \cos^2 \phi = \frac{3I}{8}$$

$$\cos^2 \phi = \frac{3}{4}$$

$$\cos^2 \phi = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \phi = 30^\circ$$

3. **Ans (543)**

$$V = 12 \text{ kV}$$

$$\text{Number of revolution} = n$$

$$n [2 \times q_p \times V] = \frac{1}{2} m_p \times v_p^2$$

$$n [2 \times 1.6 \times 10^{-19} \times 12 \times 10^3] = \frac{1}{2} \times 1.67 \times 10^{-27} \times \left[\frac{3 \times 10^8}{6} \right]^2$$

$$n(38.4 \times 10^{-16}) = 0.2087 \times 10^{-11}$$

$$n = 543.4$$

4. **Ans (14)**

$$T = 2\pi \sqrt{\frac{\ell}{g}} \Rightarrow \ell = \frac{T^2 g}{4\pi^2}$$

$$E = mg\ell \frac{\theta^2}{2} = mg^2 \frac{T^2 \theta^2}{8\pi^2}$$

$$\frac{dE}{E} = 2 \left(\frac{dg}{g} + \frac{dT}{T} \right)$$

$$= (4 + 3) = 14\%$$

5. **Ans (60)**

$$\text{Maximum emf } e = N \omega AB$$

$$N = 20, \omega = 50, B = 3 \times 10^{-2} \text{ T}$$

$$e = 20 \times 50 \times \pi \times (0.08)^2 \times 3 \times 10^{-2} = 60.28 \times 10^{-2}$$

$$\text{Rounded off to nearest integer} = 60$$

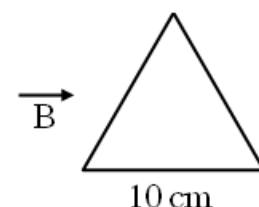
6. **Ans (2)**

$$x_2 = 5\sqrt{2} \left(\frac{1}{\sqrt{2}} \sin 2\pi t + \frac{1}{\sqrt{2}} \cos 2\pi t \right) \sqrt{2}$$

$$= 10 \sin \left(2\pi t + \frac{\pi}{4} \right)$$

$$\therefore \frac{A_2}{A_1} = \frac{10}{5} = 2$$

7. **Ans (3)**

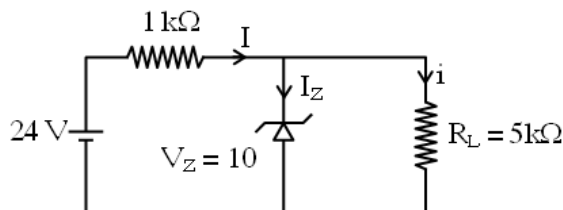


$$\vec{\tau} = \vec{M} \times \vec{B} = MB \sin 90^\circ$$

$$= MB = \frac{i\sqrt{3}\ell^2}{4} B$$

$$= \sqrt{3} \times 10^{-5} \text{ N-m}$$

8. Ans (120)



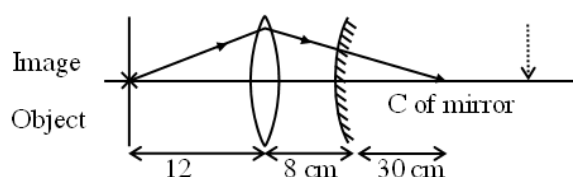
$$i = \frac{10V}{5k\Omega} = 2mA$$

$$I = \frac{14V}{1k\Omega} = 14mA$$

$$\therefore I_Z = 12mA$$

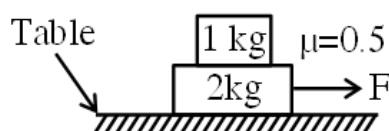
$$\therefore P = I_Z V_Z = 120 \text{ mW}$$

9. Ans (50)



For the object to coincide with image, the light must fall perpendicularly to mirror. Which means that the light will have to converge at C of mirror. Without the mirror also, the light would converge at C. So the distance is : $12 + 8 + 30 = 50 \text{ cm}$

10. Ans (15)



$$F = 3a \text{ (For system)} \quad \dots(i)$$



$$f_{s_{max}} = 1a \text{ (for 1kg block)} \quad \dots(ii)$$

$$\mu \times 1 \times g = a \Rightarrow 5 = a$$

$$F = 15N$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (B)

Only p-methyl, phenol does not give any colour with phthalic anhydride with cons. H_2SO_4 .

2. Ans (D)

Photochemical smog causes cracking of rubber, the common component of photochemical smog are ozone, nitric oxide, acrolein, formaldehyde and peroxyacetyl nitrate (PAN).

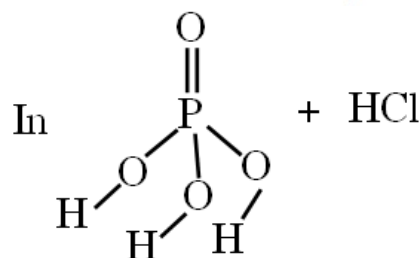
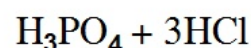
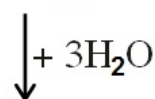
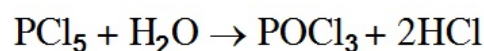
3. Ans (C)

For london dispersion forces.

$$E \propto \frac{1}{r^6}$$

$$\text{Hence } x = -6$$

4. Ans (A)



All hydrogens are ionisable

$\therefore$ Ans is zero.

5. Ans (A)

$$O_2^- = (\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\sigma_{2p_z})^2$$

$$\left(\pi_{2p_x}^2 = \pi_{2p_y}^2 \right) \left(\pi_{2p_x}^{*2} = \pi_{2p_y}^{*1} \right)$$

$$\text{Bond order} = \frac{10 - 7}{2} = 1.5$$

and paramagnetic

6. Ans (C)

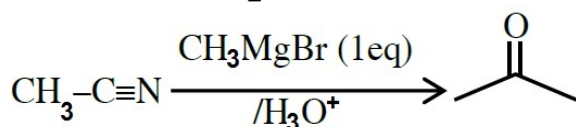
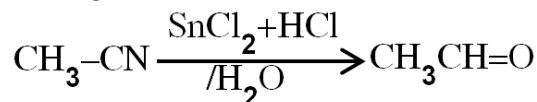
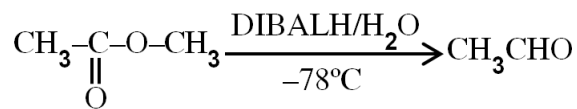
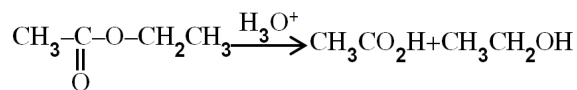
Sucrose is example of disaccharide & non reducing sugar

Assertion : correct

Sucrose involves glycosidic linkage between C_1 of α -D-glucose C_2 of β -D-fructose

Reason : Incorrect

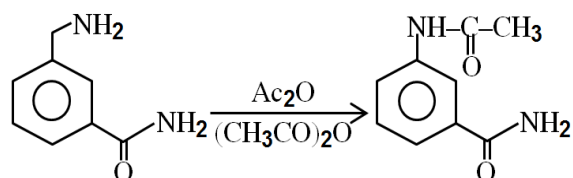
7. **Ans (C)**



8. **Ans (A)**

In IIA group on moving down the group size of cation increases and show thermal stability of carbonate increases.

9. **Ans (D)**



10. **Ans (B)**

(1) $[\text{CoCl}_2(\text{en})_2]$ show

Cis-trans isomerism

(2) $[\text{Co}(\text{CN})_5(\text{NC})]^{-3}$ can't

Show G.I.

(3) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$

Show fac & mer isomerism

(4) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^{\oplus}$ show cis & trans isomerism

12. **Ans (B)**

Sphalerite-ZnS, copper glance - Cu_2S two sulphide ores can be separated by adjusting proportions of oil to water or by using 'Depressants'

13. **Ans (D)**

D_2O is used for the study of reaction mechanism. Rate of reaction for the cleavage of O-H bond > O-D bond.

14. **Ans (B)**

(i) $\text{CFSE} \propto$ charge or oxidation no. of central metal ion.

(ii) $\text{CFSE} \propto$ strength of ligand

$\text{en} > \text{NH}_3 > \text{H}_2\text{O} > \text{F}^-$

$\therefore$ order of CFSE

$[\text{Co}(\text{en})_3]^{+3} > [\text{Co}(\text{NH}_3)_6]^{+3} > [\text{CoF}_6]^{-3} > [\text{Co}(\text{H}_2\text{O})_6]^{+2}$

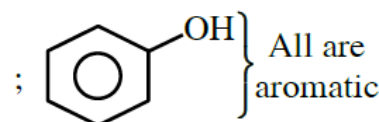
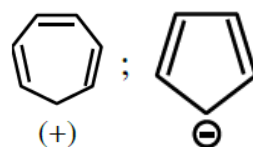
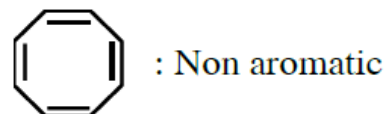
15. **Ans (C)**

The drug named chlordiazepoxide is example of tranquilizer.

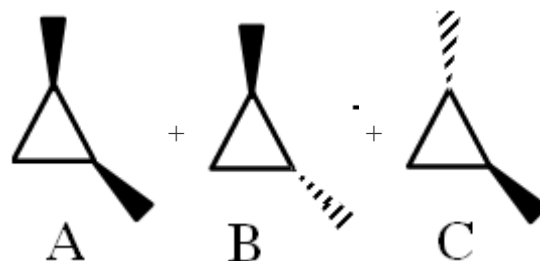
16. **Ans (B)**

Group 16/oxygen family is known as Chalcogens the members are O, S, Se, Te, Po

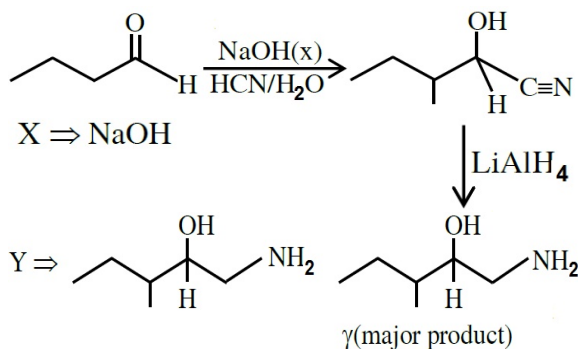
17. **Ans (C)**



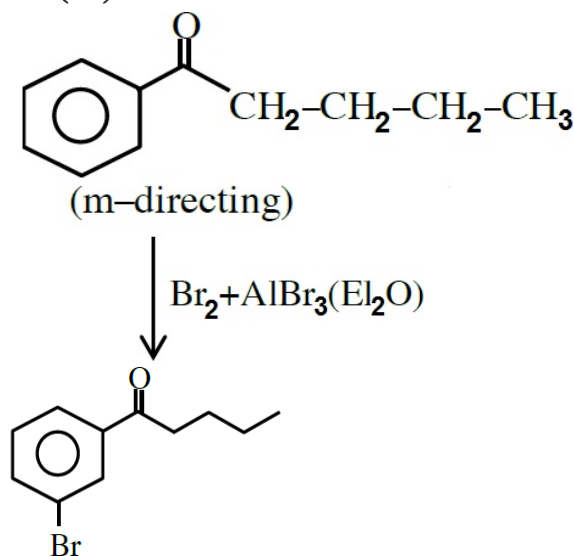
18. **Ans (D)**



19. Ans (C)



20. Ans (C)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (42)

Molecular mass of $\text{BaSO}_4 = 233 \text{ g}$

$\therefore 233 \text{ BaSO}_4$ contain $\rightarrow 32 \text{ g}$ sulphur

$\therefore 1.44 \text{ g BaSO}_4$ contain $\rightarrow \frac{32}{233} \times 1.44 \text{ g}$

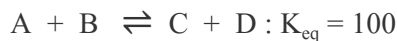
sulphur

given : 0.471 g of organic compound

$$\% \text{ of S} = \frac{32 \times 1.44}{233 \times 0.471} \times 100$$

$$= 41.98\% \approx 42\%$$

2. Ans (182)



1M 1M 1M 1M

First check direction of reversible reaction.

Since $Q_c =$

$$\frac{[C][D]}{[A][B]} = 1 < K_{\text{eq}} \Rightarrow \text{reaction will move in}$$

forward direction to attain equilibrium state.



to 1 1 1 1

$t_{\text{eq.}} 1-x 1-x 1+x 1+x$

$$\text{Now : } K_{\text{eq}} = 100 = \frac{(1+x)(1+x)}{(1-x)(1-x)}$$

$$\Rightarrow 100 = \left(\frac{1+x}{1-x} \right)^2$$

$$(i) 10 = \left(\frac{1+x}{1-x} \right)$$

$$\Rightarrow 10 - 10x = 1+x$$

$$\Rightarrow 11x = 9$$

$$\Rightarrow x = \frac{9}{11}$$

$$(ii) -10 = \frac{1+x}{1-x}$$

$$\Rightarrow -10 + 10x = 1+x$$

$$\Rightarrow -9x = -11$$

$$\Rightarrow x = \frac{11}{9}$$

$\rightarrow$ 'x' cannot be more than one, therefore not valid.

therefore equation concretion of (D) = $1+x$

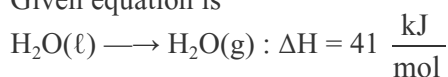
$$= 1 + \frac{9}{11} = \frac{20}{11}$$

$$= 1.8181 = 181.81 \times 10^{-2}$$

$$= 182 \times 10^{-2}$$

3. Ans (38)

Given equation is



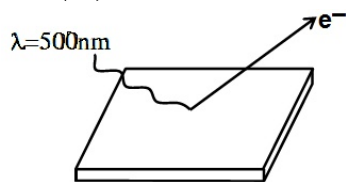
$\Rightarrow$ From the relation : $\Delta H = \Delta U + \Delta n_g RT$

$$\Rightarrow 41 \frac{\text{kJ}}{\text{mol}} = \Delta U + (1) \times \frac{8.3}{1000} \times 373$$

$$\Rightarrow \Delta U = 41 - 3.0959$$

$$= 38 \text{ kJ/mol}$$

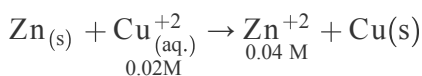
4. Ans (5)



v : speed of electron having max. K.E.
 $\Rightarrow$ from Einstein equation : $E = \phi + K.E._{max}$
 $\Rightarrow \frac{hc}{\lambda} = h\nu_0 + \frac{1}{2}mv^2$
 $\Rightarrow \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{500 \times 10^{-9}}$
 $= 6.63 \times 10^{-34} \times 4.3 \times 10^{14} + \frac{1}{2}mv^2$
 $\Rightarrow \frac{6.63 \times 30 \times 10^{-20}}{5}$
 $= 6.63 \times 4.3 \times 10^{-20} + \frac{1}{2}mv^2$
 $\Rightarrow 11.271 \times 10^{-20} \text{ J} = \frac{1}{2} \times 9 \times 10^{-31} \times v^2$
 $\Rightarrow \boxed{v = 5 \times 10^5 \text{ m/sec.}}$

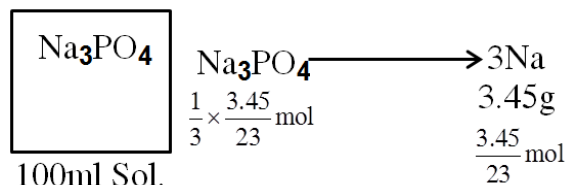
5. Ans (109)

Galvanic cell:



Nernst equation = $E_{cell} = E^\circ_{cell} - \frac{0.059}{2} \log \frac{[2n^{+2}]}{[Cu^{+2}]}$
 $\Rightarrow E_{cell} \left[E^\circ_{cell} - E^\circ_{Zn^{+2}/Zn} \right] - \frac{0.059}{2} \log \frac{0.04}{0.02}$
 $\Rightarrow E_{cell} [0.34 - (-0.76)] - \frac{0.059}{2} \log^2$
 $\Rightarrow E_{cell} 1 - 1 - \frac{0.059}{2} \times 0.3010$
 $= 1.0911 = 109.11 \times 10^{-2}$
 $= 109$

6. Ans (50)



100ml Sol.

therefore molarity of Na_3PO_4 Solution =

$$\frac{n_{\text{Na}_3\text{PO}_4}}{\text{volume of solution in L}}$$

 $= \frac{\frac{1}{3} \times \frac{3.45}{23} \text{ mol}}{0.1 \text{ L}}$
 $= 0.5 = 50 \times 10^{-2}$

7. Ans (5)

Given $k_f = 2.1 \times 10^{13}$
 $K_d = \frac{1}{K_f} = 4.7 \times 10^{-14}$

$\therefore y = 4.7 \approx 5$

8. Ans (269)

$k_f = 1.86 \text{ k. kg/mol}$

$T_f^\circ = 273 \text{ K}$

solvent : H_2O (625 g)
 Solute : 83 g $\left(\begin{array}{cc} \text{CH}_2 - \text{CH}_2 \\ | \quad \quad | \\ \text{OH} \quad \quad \text{OH} \end{array} \right) \Rightarrow$

Non dissociative solute

$\Rightarrow \Delta T_f = k_f \times m$

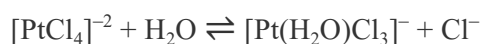
$\Rightarrow (T_f^\circ - T_f^1) = 1.86 \times \frac{83/62}{624/1000} \Rightarrow 273 -$

$T_f^1 = \frac{1.86 \times 83 \times 1000}{62 \times 625} = \frac{154380}{38750}$

$\Rightarrow 273 - T_f^1 = 4$

$\Rightarrow \boxed{T_f^1 = 259 \text{ K}}$

9. Ans (0)



$$\frac{-d[\text{PtCl}_4]^{-2}}{dt} = 4.8 \times 10^{-5} [\text{PtCl}_4^{-2}] - 2.4 \times$$

$10^3 [\text{Pt}(\text{H}_2\text{O})\text{Cl}_3][\text{u}]$

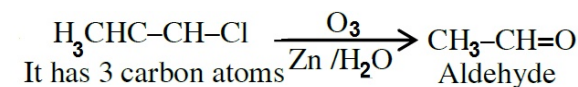
$\Rightarrow K_{eq} = \frac{k_f}{k_b} = \frac{4.8 \times 10^{-5}}{2.4 \times 10^{-3}} = 0.02$

10. Ans (3)

448 ml of A $\Rightarrow 1.53 \text{ gm A}$

22400 ml of A $\Rightarrow \frac{1.53}{445} \times 22400 \text{ gm A}$

$= 7650$



& mm is $36 + 5 + 35.5 = 76.5$

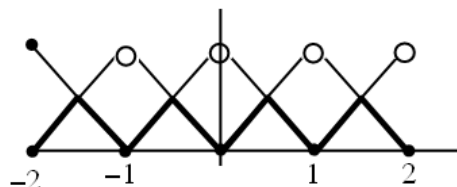
PART-3 : MATHEMATICS

SECTION-I

1. Ans (A)

$$\min\{x - [x], 1 - x + [x]\}$$

$$h(x) = \min\{x - [x], 1 - [x - [x]]\}$$



$\Rightarrow$ always continuous in $[-2, 2]$

but non differentiable at 7 Points

2. Ans (A)

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A^n = \begin{bmatrix} 1 & 0 & 0 \\ n-1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A^{2025} - A^{2020} = \begin{bmatrix} 0 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^6 - A = \begin{bmatrix} 0 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3. Ans (C)

$$f(x) = \left(\frac{2}{x}\right)^{x^2}; x > 0$$

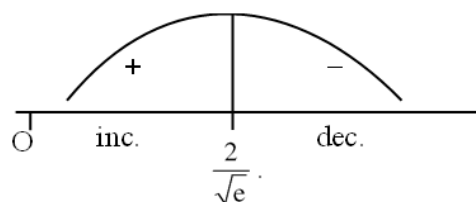
$$\ln f(x) = x^2 (\ln 2 - \ln x)$$

$$f'(x) = f(x) \{-x + (\ln 2 - \ln x)2x\}$$

$$f'(x) = \underbrace{f(x)}_{+} \cdot \underbrace{x}_{+} \underbrace{(2\ln 2 - 2\ln x - 1)}_{g(x)}$$

$$g(x) = 2\ln 2 - 2\ln x - 1$$

$$= \ln \frac{4}{x^2} - 1 = 0 \Rightarrow x = \frac{2}{\sqrt{e}}$$



$$LM = \frac{2}{\sqrt{e}}$$

$$\text{Local maximum value} = \left(\frac{2}{2/\sqrt{e}}\right)^{\frac{4}{e}} \Rightarrow e^{\frac{2}{e}}$$

4. Ans (B)

$$I = \int_0^5 \frac{x + [x]}{e^{x-[x]}} dx$$

$$\int_0^1 \frac{x}{e^x} dx + \int_1^2 \frac{x+1}{e^{x-1}} dx + \int_2^3 \frac{x+2}{e^{x-2}} dx + \dots + \int_4^5 \frac{x+4}{e^{x-4}} dx$$

$\downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow$
 $x = t + 1 \qquad x = z + 2 \qquad x = y + 4$

$$\int_0^1 \frac{t+2}{e^t} dt + \int_0^1 \frac{z+4}{e^z} dz + \dots + \int_0^1 \frac{y+8}{e^y} dy$$

$$\Rightarrow \int_0^5 \frac{5x+20}{e^x} dx = 5 \int_0^1 \frac{x+4}{e^x} dx$$

$$\Rightarrow 5 \int_0^1 (x+4)e^{-x} dx$$

$$\Rightarrow 5 e^{-x} (-x-5) \Big|_0^1 \Rightarrow -\frac{30}{e} + 25$$

$$\alpha = -30$$

$$\beta = 25 \Rightarrow 5\alpha + 6\beta = 0$$

$$(\alpha + \beta)^2 = 5^2 = 25$$

5. **Ans (C)**

$P(-2\sqrt{6}, \sqrt{3})$ lies on hyperbola

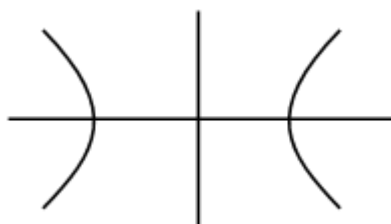
$$\Rightarrow \frac{24}{a^2} - \frac{3}{b^2} = 1 \quad \dots\dots(i)$$

$$e = \frac{\sqrt{5}}{2} \Rightarrow b^2 = a^2 \left(\frac{5}{4} - 1 \right) \Rightarrow 4b^2 = a^2$$

$$\text{Put in (i)} \Rightarrow \frac{6}{b^2} - \frac{3}{b^2} = 1 \Rightarrow b = \sqrt{3}$$

$$\Rightarrow a = \sqrt{12}$$

$$\frac{x^2}{12} - \frac{y^2}{3} = 1$$



Tangent at P :

$$\frac{-x}{\sqrt{6}} - \frac{y}{\sqrt{3}} = 1 \Rightarrow Q(0, \sqrt{3})$$

$$\text{Slope of T} = -\frac{1}{\sqrt{2}}$$

$$\text{Normal at P : } y - \sqrt{3} = \sqrt{2}(x + 2\sqrt{6})$$

$$\Rightarrow R = (0, 5\sqrt{3})$$

$$QR = 6\sqrt{3}$$

6. **Ans (C)**

$$2x^2 dy + (e^y - 2x) dx = 0$$

$$\frac{dy}{dx} + \frac{e^y - 2x}{2x^2} = 0 \Rightarrow$$

$$\frac{dy}{dx} + \frac{e^y}{2x^2} - \frac{1}{x} = 0$$

$$e^{-y} \frac{dy}{dx} - \frac{e^{-y}}{x} = -\frac{1}{2x^2} \Rightarrow \text{Put } e^{-y} = z$$

$$\frac{-dz}{dx} - \frac{z}{x} = -\frac{1}{2x^2} \Rightarrow xdz + zdx = \frac{dx}{2x}$$

$$d(xz) = \frac{dx}{2x} \Rightarrow xz = \frac{1}{2} \log_e x + c$$

$$xe^{-y} = \frac{1}{2} \log_e x + c, \text{ passes through } (e, 1)$$

$$\Rightarrow C = \frac{1}{2}$$

$$xe^{-y} = \frac{\log_e x}{2}$$

$$e^{-y} = \frac{1}{2} \Rightarrow y = \log_e 2$$

7. **Ans (C)**

$$S_1 : (\sim p \vee q) \vee (q \vee p) = (q \vee \sim p) \vee (q \vee p)$$

$$S_1 = q \vee (\sim p \vee p) = q \vee t = t = \text{tautology}$$

$$S_2 : (p \wedge \sim q) \wedge (\sim p \vee q) = (p \wedge \sim q) \wedge \sim (p \wedge \sim q) = C$$

$$= \text{fallacy}$$

8. **Ans (D)**

$$\frac{1+x}{x} \in (-\infty, -1] \cup [1, \infty)$$

$$\frac{1}{x} \in (-\infty, -2] \cup [0, \infty)$$

$$x \in \left[-\frac{1}{2}, 0\right) \cup (0, \infty)$$

$$x \in \left[-\frac{1}{2}, \infty\right) - \{0\}$$

9. **Ans (D)**

$$P(x \geq 5 | x > 2) = \frac{P(x \geq 5)}{P(x > 2)}$$

$$\frac{\left(\frac{5}{6}\right)^4 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^5 \cdot \frac{1}{6} + \dots + \infty}{\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^3 \cdot \frac{1}{6} + \dots + \infty}$$

$$\frac{\left(\frac{5}{6}\right)^4 \cdot \frac{1}{6}}{\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6}}$$

$$\frac{1 - \frac{5}{6}}{1 - \frac{5}{6}} = \left(\frac{5}{6}\right)^2 = \frac{25}{36}$$

10. **Ans (B)**

$$\sum_{r=1}^{50} \tan^{-1} \left(\frac{2}{4r^2} \right) = \sum_{r=1}^{50} \tan^{-1} \left(\frac{(2r+1) - (2r-1)}{1 + (2r+1)(2r-1)} \right)$$

$$\sum_{r=1}^{50} \tan^{-1} (2r+1) - \tan^{-1} (2r-1)$$

$$\tan^{-1} (101) - \tan^{-1} 1 \Rightarrow \tan^{-1} \frac{50}{51}$$

11. Ans (B)

$$D \neq 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & \lambda \end{vmatrix} \neq 0 \Rightarrow \lambda \neq 5$$

For no solution $D = 0 \Rightarrow \lambda = 5$

$$D_1 = \begin{vmatrix} 1 & 1 & 5 \\ 1 & 2 & \mu \\ 1 & 3 & 1 \end{vmatrix} \neq 0 \Rightarrow \mu \neq 3$$

$$p = \frac{5}{6}$$

$$q = \frac{1}{6} \times \frac{5}{6} = \frac{5}{36}$$

12. Ans (C)

$$T = S_1$$

$$xh - yk = h^2 - k^2 y = \frac{xh}{k} - \frac{(h^2 - k^2)}{k}$$

this touches $y^2 = 8x$ then $c = \frac{a}{m}$

$$\left(\frac{k^2 - h^2}{k} \right) = \frac{2k}{h}$$

$$2y^2 = x(y^2 - x^2)$$

$$y^2(x - 2) = x^3$$

13. Ans (C)

$$2 \sin\left(\frac{\pi}{8}\right) \sin\left(\frac{2\pi}{8}\right) \sin\left(\frac{3\pi}{8}\right) \sin\left(\frac{5\pi}{8}\right) \sin\left(\frac{6\pi}{8}\right) \sin\left(\frac{7\pi}{8}\right)$$

$$2 \sin^2 \frac{\pi}{8} \sin^2 \frac{2\pi}{8} \sin^2 \frac{3\pi}{8}$$

$$\sin^2 \frac{\pi}{8} \sin^2 \frac{3\pi}{8}$$

$$\sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8}$$

$$\frac{1}{4} \sin^2 \left(\frac{\pi}{4} \right) = \frac{1}{8}$$

14. Ans (A)

$$\left(2e^{i\pi/6} \right)^{100} = 2^{99} (p + iq)$$

$$2^{100} \left(\cos \frac{50\pi}{3} + i \sin \frac{50\pi}{3} \right) = 2^{99} (p + iq)$$

$$p + iq = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$p = -1, q = \sqrt{3}$$

$$x^2 - (\sqrt{3} - 1)x - \sqrt{3} = 0$$

15. Ans (D)

$$A(\hat{j}), B(10\hat{i})$$

$$H(h\hat{j} + 10\hat{k})$$

$$G(10\hat{i} + h\hat{j} + 10\hat{k})$$

$$\vec{AG} = 10\hat{i} + h\hat{j} + 10\hat{k}$$

$$\vec{BH} = -10\hat{i} + h\hat{j} + 10\hat{k}$$

$$\cos \theta = \frac{\vec{AG} \cdot \vec{BH}}{|\vec{AG}| |\vec{BH}|}$$

$$\frac{1}{5} = \frac{h^2}{h^2 + 200}$$

$$4h^2 = 200 \Rightarrow h = 5\sqrt{2}$$

16. Ans (C)

$$(x + y + 4z - 16) + \lambda(-x + y + z - 6) = 0$$

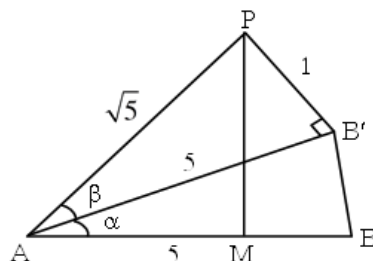
Passes through (1, 2, 3)

$$-1 + \lambda(-2) \Rightarrow \lambda = -\frac{1}{2}$$

$$2(x + y + 4z - 16) - (-x + y + z - 6) = 0$$

$$3x + y + 7z - 26 = 0$$

17. Ans (A)



From figure

$$\sin \beta = \frac{1}{\sqrt{5}}$$

$$\therefore \tan \beta = \frac{1}{2}$$

$$\tan(\alpha + \beta) = 2$$

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} = 2$$

$$\frac{\tan \alpha + \frac{1}{2}}{1 - \tan \alpha \left(\frac{1}{2} \right)} = 2$$

$$\tan \alpha = \frac{3}{4}$$

$$\alpha = \tan^{-1} \left(\frac{3}{4} \right)$$

18. Ans (C)

$$I = \int_0^{\pi/2} \frac{(1 + \sin^2 x)}{(1 + \pi \sin x)} + \frac{\pi \sin x (1 + \sin^2 x)}{(1 + \pi \sin x)} dx$$

$$I = \int_0^{\pi/2} (1 + \sin^2 x) dx$$

$$I = \frac{\pi}{2} + \frac{\pi}{2} \cdot \frac{1}{2} = \frac{3\pi}{4}$$

19. Ans (A)

$$(x-2)^2 + (y-1)^2 + \lambda(x-2y) = 0$$

$$C : x^2 + y^2 + x(\lambda-4) + y(-2-2\lambda) + 5 = 0$$

$$C_1 : x^2 + y^2 + 2y - 5 = 0$$

$$S_1 - S_2 = 0 \text{ (Equation of PQ)}$$

$$(\lambda-4)x - (2\lambda+4)y + 10 = 0 \text{ Passes through } (0, -1)$$

$$\Rightarrow \lambda = -7$$

$$C : x^2 + y^2 - 11x + 12y + 5 = 0$$

$$= \frac{\sqrt{245}}{4}$$

$$\text{Diameter} = 7\sqrt{5}$$

20. Ans (A)

$$S = \lim_{x \rightarrow 2} \sum_{n=1}^9 \frac{x}{n(n+1)x^2 + 2(2n+1)x + 4}$$

$$S = \sum_{n=1}^9 \frac{2}{4(n^2 + 3n + 2)} = \frac{1}{2} \sum_{n=1}^9 \left(\frac{1}{n+1} - \frac{1}{n+2} \right)$$

$$S = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{11} \right) = \frac{9}{44}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (7744)

$$209, 220, 231, \dots, 495$$

$$\text{Sum} = \frac{27}{2} (209 + 495) = 9504$$

$$\underline{2} \quad \underline{3} \quad \underline{1}$$

$$\text{Number containing 1 at unit place} \quad \underline{3} \quad \underline{4} \quad \underline{1}$$

$$\underline{4} \quad \underline{5} \quad \underline{1}$$

$$\underline{3} \quad \underline{1} \quad \underline{9}$$

$$\text{Number containing 1 at } 10^{\text{th}} \text{ place}$$

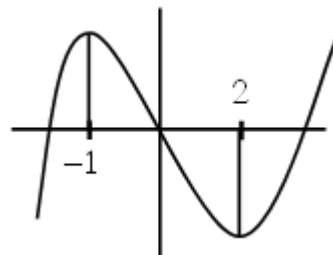
$$\underline{4} \quad \underline{1} \quad \underline{8}$$

$$\text{Required} = 9501 - (231 + 341 + 451 + 319 + 418) = 7744$$

2. Ans (114)

$$f(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1)$$

$$\text{Point} = (2, -20) \text{ \& } (-1, 7)$$



$$A = \int_{-1}^0 (2x^3 - 3x^2 - 12x) dx + \int_0^2 (12x + 3x^2 - 2x^3) dx$$

$$A = \left(\frac{x^4}{2} - x^3 - 6x^2 \right)_{-1}^0 + \left(6x^2 + x^3 - \frac{x^4}{2} \right)_0^2$$

$$4A = 114$$

3. Ans (5)

$$\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$$

$$\vec{b} = (2-\lambda)\hat{i} + 6\hat{j} - 2\hat{k}$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = 1, \quad \vec{a} \cdot \vec{b} = 12 - \lambda$$

$$(\vec{a} \cdot \vec{b}) = |\vec{b}|^2$$

$$\lambda^2 - 24\lambda + 144 = \lambda^2 - 4\lambda + 4 + 40$$

$$20\lambda = 100 \Rightarrow \lambda = 5$$

4. Ans (2021)

$$c_2 = a_2 + b_2 = a_1 - 3 + 2b_1 = 12$$

$$a_1 + 2b_1 = 15 \quad (1)$$

$$c_3 = a_3 + b_3 = a_1 - 6 + 4b_1 = 13$$

$$a_1 + 4b_1 = 19 \quad (2)$$

$$\text{from (1) \& (2) } b_1 = 2, a_1 = 11$$

$$\sum_{k=1}^{10} c_k = \sum_{k=1}^{10} (a_k + b_k) = \sum_{k=1}^{10} a_k + \sum_{k=1}^{10} b_k$$

$$= \frac{10}{2} (2 \times 11 + 9 \times (-3)) + \frac{2(2^{10} - 1)}{2 - 1}$$

$$= 5(22 - 27) + 2(1023)$$

$$= 2046 - 25 = 2021$$

5. Ans (96)

Containing the line

$$\begin{vmatrix} x+1 & y-1 & z-3 \\ 6 & 7 & 8 \\ 3 & 5 & 7 \end{vmatrix} = 0$$

$$9(x+1) - 18(y-1) + 9(z-3) = 0$$

$$x - 2y + z = 0$$

$$PQ = \left| \frac{7+4+13}{\sqrt{6}} \right| = 4\sqrt{6}$$

$$PQ^2 = 96$$

6. Ans (49)

$$A_k = \sum_{i=0}^9 {}^9C_i {}^{12}C_{k-i} + \sum_{i=0}^8 {}^8C_i {}^{13}C_{k-i}$$

$$A_k = {}^{21}C_k + {}^{21}C_k = 2 \cdot {}^{21}C_k$$

$$A_4 - A_3 = 2({}^{21}C_4 - {}^{21}C_3) = 2(5985 - 1330)$$

$$190p = 2(5985 - 1330) \Rightarrow p = 49$$

7. Ans (18)

$$3\alpha^2 - 10\alpha + 27\lambda = 0 \quad \text{---(1)}$$

$$\alpha^2 - \alpha + 2\lambda = 0 \quad \text{---(2)}$$

(1) - 3(2) gives

$$-7\alpha + 21\lambda = 0 \Rightarrow \alpha = 3\lambda$$

Put $\alpha = 3\lambda$ in equation (1) we get

$$9\lambda^2 - 3\lambda + 2\lambda = 0$$

$$9\lambda^2 = \lambda \Rightarrow \lambda = \frac{1}{9} \text{ as } \lambda \neq 0$$

$$\text{Now } \alpha = 3\lambda \Rightarrow \lambda = \frac{1}{3}$$

$$\alpha + \beta = 1 \Rightarrow \beta = 2/3$$

$$\alpha + \gamma = \frac{10}{3} \Rightarrow \gamma = 3$$

$$\frac{\beta\gamma}{\lambda} = \frac{\frac{2}{3} \times 3}{\frac{1}{9}} = 18$$

8. Ans (12)

$$5 = \frac{3+7+x+y}{4} \Rightarrow x+y = 10$$

$$\text{Var}(x) = 10 = \frac{3^2 + 7^2 + x^2 + y^2}{4} - 25$$

$$140 = 49 + 9 + x^2 + y^2$$

$$x^2 + y^2 = 82$$

$$x + y = 10$$

$$\Rightarrow (x, y) = (9, 1)$$

Four numbers are 21, 9, 10, 8

$$\text{Mean} = \frac{48}{4} = 12$$

9. Ans (4)

$$\text{adj}(2A) = 2^2 \text{adj}A$$

$$\Rightarrow \text{adj}(\text{adj}(2A)) = \text{adj}(4 \text{adj}A) = 16 \text{adj}(\text{adj}A)$$

$$= 16 |A| A$$

$$\Rightarrow \text{adj}(32 |A| A) = (32 |A|)^2 \text{adj}A$$

$$12(32|A|)^2 |\text{adj}A| = 2^3 (32|A|)^6 |\text{adj}A|$$

$$2^3 \cdot 2^{30} |A|^6 \cdot |A|^2 = 2^{41}$$

$$|A|^8 = 2^8 \Rightarrow |A| = \pm 2$$

$$|A|^2 = |A|^2 = 4$$

10. Ans (6)

$$\begin{aligned} \frac{(2i)^n}{(1-i)^{n-2}} &= \frac{(2i)^n}{(-2i)^{\frac{n-2}{2}}} \\ &= \frac{(2i)^{\frac{n+2}{2}}}{(-1)^{\frac{n-2}{2}}} = \frac{2^{\frac{n+2}{2}} i^{\frac{n+2}{2}}}{(-1)^{\frac{n-2}{2}}} \end{aligned}$$

This is positive integer for $n = 6$